

# RC 2nd Year Interview Syllabus

You need to prepare for your chosen domain.

## Mechanical:-

1. Parametric & Sheet Metal CAD: Tolerancing for type of manufacturing, Types of Fit, Strategic sheet metal bends and flat patterns; Stress Concentration and part lighting, modelling and assembly best practices in any of Fusion360, Onshape, Solidworks.
2. Kinematics & Mechanisms: Various ways of achieving Linear and Rotary motion, calculating DoF and how to constrain motion; Rules for Four-Bar Linkages, Coaxial virtual 4 bars, Ratchet & Pawl, Differentials mechanism.
3. Manufacturing and Miscellaneous: Difference between Strength and Stiffness, Types of joining methods, Welding and Base Materials; Explaining Turning, Milling, Drilling, Tapping, Popular FDM 3D printing materials, their warping and use cases, how to increase traction on wheels, Spring Calculation, Damping, counterbalancing.
4. Actuators and Torque Transfer: DC motor Torque-Speed Curves, Peak power vs Peak torque point, how PWM affects DC motor performance and heating, Sizes of DC motors available in the market and their shaft types; Types of Gears, belts, chains, and their use cases, High torque motors vs High ratio gearboxes.

## Embedded:-

1. Types of Sensors and Actuators and how to interface with them: Pros and Cons of I2C, PWM, SPI, UART, CAN. Difference between DC, BLDC, Servo and Stepper Motors in regards to their use and Torque-Speed Curves, how PWM affects DC motor performance and heating. Motor Breaking vs Coasting, Open loop vs Closed loop control.
2. C++ or Python, ESP32 features, Vin and max out current limitations, Logic Level, Using Arduino Libraries and Interrupt Service Routines, Basics of finite state machines and control systems, Cascade PID, Proportional on error vs on measurement, Feed forward vs Feedback.

3. Resistors (pull-up and pull-down), capacitors (decoupling), transistors (use of them as switches), relays, hardware and software debouncing, Emergency stop wiring, how to mitigate ESD, proper grounding technique, most popular types of wire connectors.

## Software:-

- **ROS2:** Program, coordinate, and manage modular robot software using nodes, topics, and actions.
- **Simulation tools (Gazebo, Coppeliasim):** Practice building, testing, and visualizing robots in realistic virtual environments.
- **ML/OpenCV:** Master computer vision, image analysis, and applying machine learning models for perception tasks.
- **OS:** Understand operating system principles, process management, and hardware-software interfacing.
- **SQL:** Learn to create, query, and maintain relational databases efficiently.
- **WebDev:** Gain skills to design, implement, and deploy interactive websites and web applications.